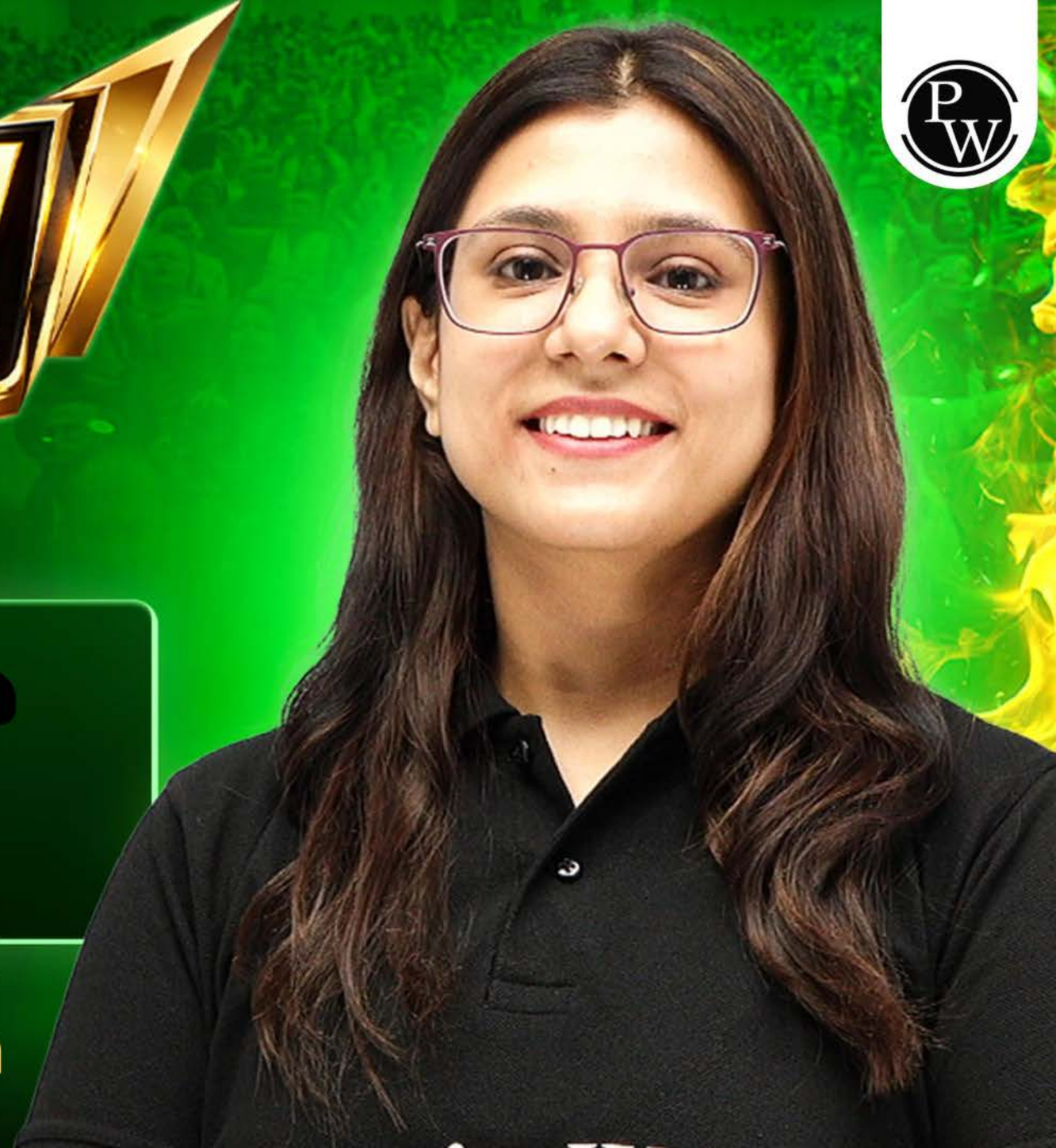
VICTORY 2027

THE CIRCULATORY SYSTEM

LECTURE-01 | BIOLOGY

By– Upasana Ma'am



TOPICS TO BE COVERED

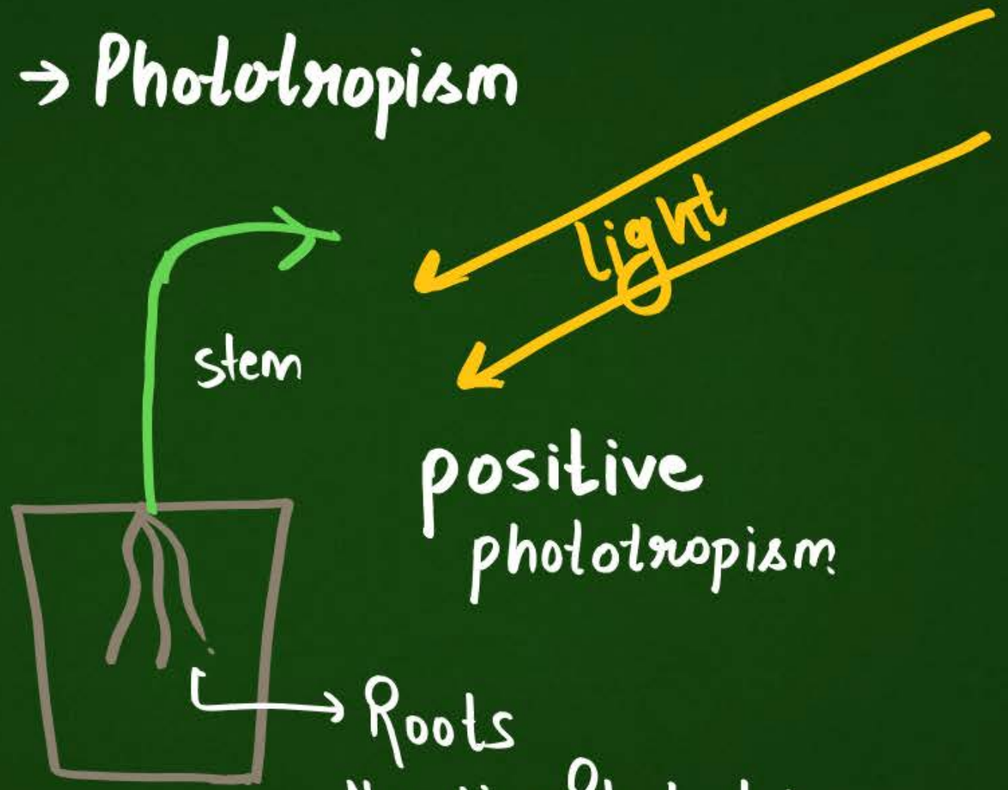
TROPIC MOVEMENT IN PLANTS

CIRCULATORY SYSTEM- Components

Circulation Introduction

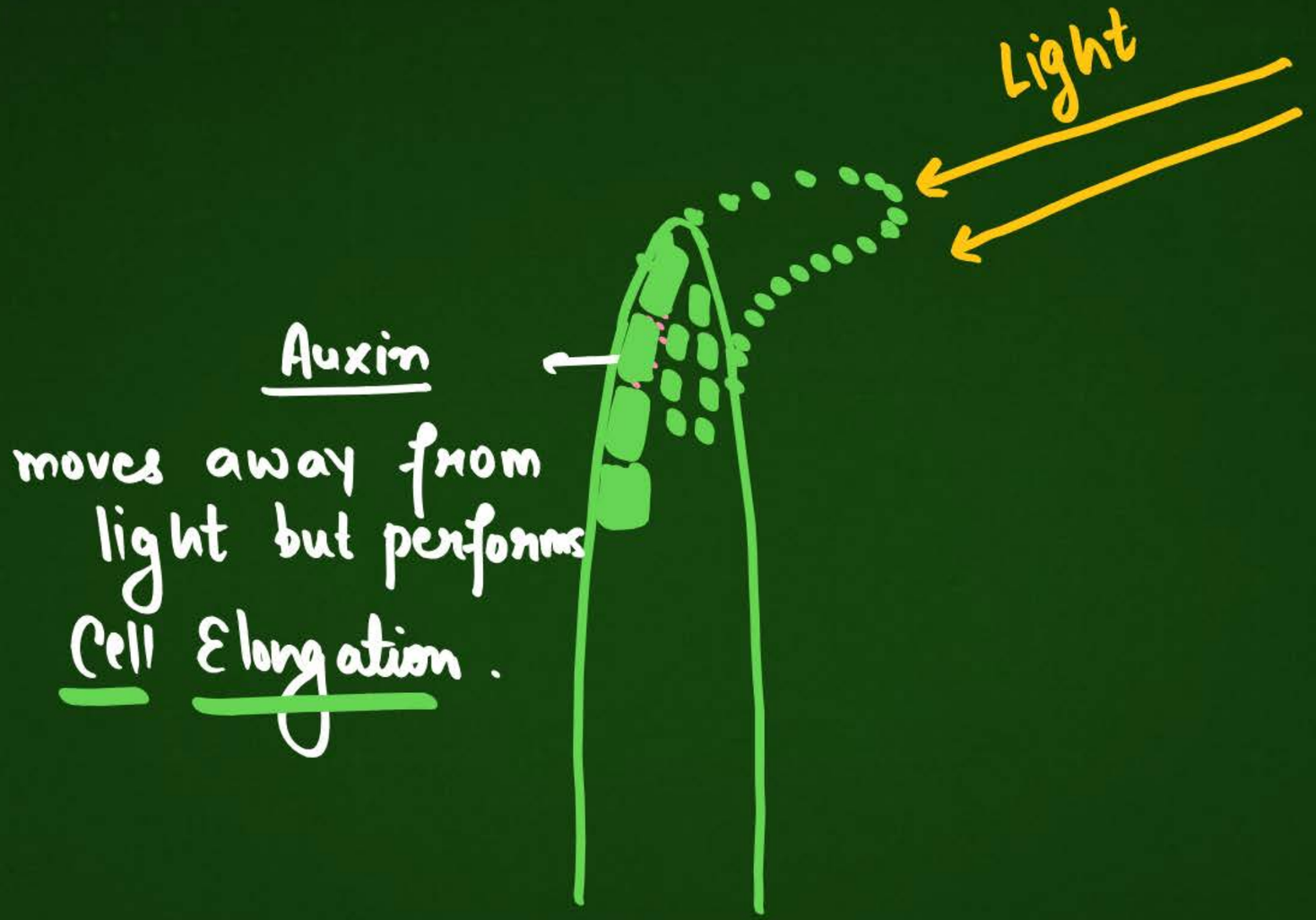

HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

light → Phototropism



positive phototropism

Negative Photropism.


Light
Auxin
moves away from light but performs Cell Elongation.

SECTION B (40 Marks)

*(Attempt **any four** questions from this **Section.**)*

Question 3

(i) Write the overall chemical equation for photosynthesis. [1]

(ii) Ramesh is a farmer who has a vineyard and cultivates grapes. To improve the shape of the grapes and make them elongated he sprays a particular *phytohormone* at the time of fruiting. [2]



(a) Name the *phytohormone* that Ramesh sprays on the plant at the time of fruiting.

(b) These grapes are said to be seedless. What is the technical term for the formation of such fruits?

Plant hormone	Site of synthesis	Functions
1. Auxins	• Shoot apical meristems and young leaves	• Promote cell elongation Suppress the growth of lateral buds Delay fall of leaves Induce formation of parthenocarpic fruits.
2. Gibberellins	• Meristem of apical buds and roots	• Help in stem elongation Break dormancy of seeds and buds Delay senescence Induce parthenocarpy
3. Cytokinin	• Roots and transported to other organs	• Stimulate cell division and cell enlargement Prevent ageing of plant parts Inhibit apical dominance
4. Ethylene	• Senescent leaves and flowers, germinating seeds and ripening fruits.	• Induces fruit ripening Promote senescence
5. Abscisic acid	• Green fruits and seeds at the beginning of wintering periods.	• Induces dormancy of buds and seeds Inhibits seed germination and development Stimulates closing of stomata

Tropic movements in plants

- The root must grow downwards into soil which will provide support, water and minerals for the plants.
- Shoot must grow up towards the light source so that developing leaflets could carry out photosynthesis in order to produce food for the growing plant.

Tropic movements in plants

- Movement of these parts of the plant take place in direct response to external stimuli.
- The direction of response is related to direction from which stimulus comes. Such response is known as tropism.
- Growth movements occurring in response to unidirectional external stimuli in a plant part are called tropic movements.

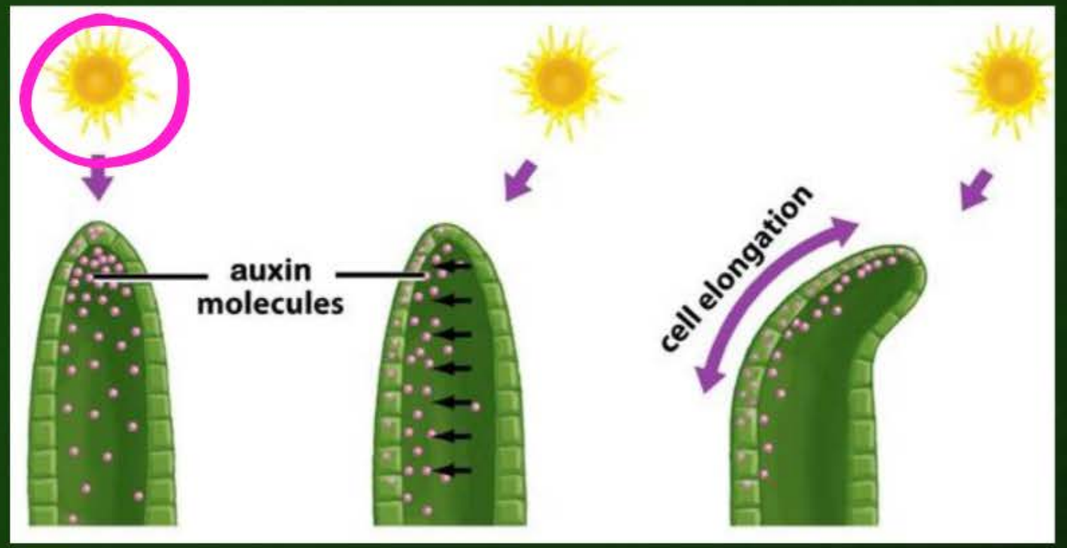
I. Phototropism

Movement towards light

- Shoots are positively phototropic, and roots are negatively phototropic

Auxins play an important role

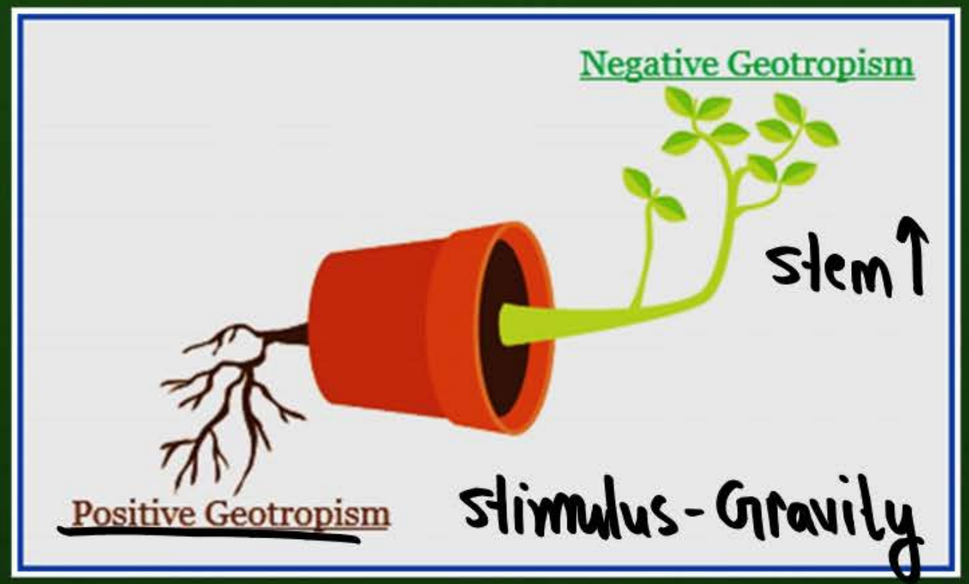
- Auxins help in bending of shoot towards the source of light by getting accumulated in the region of the shoot which is not facing light which result in rapid cell-elongation.



II. Geotropism

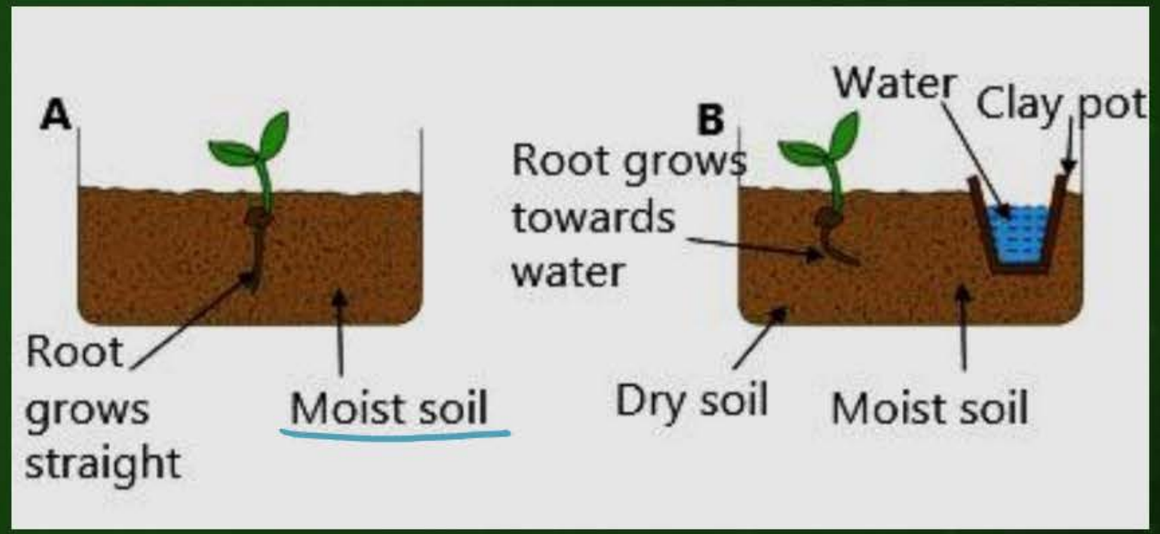
Geo - Earth

- Growing towards the earth's gravity
- Also known as gravitropism.
- Roots tip are positively geotropic and shoot tip is negatively geotropic.

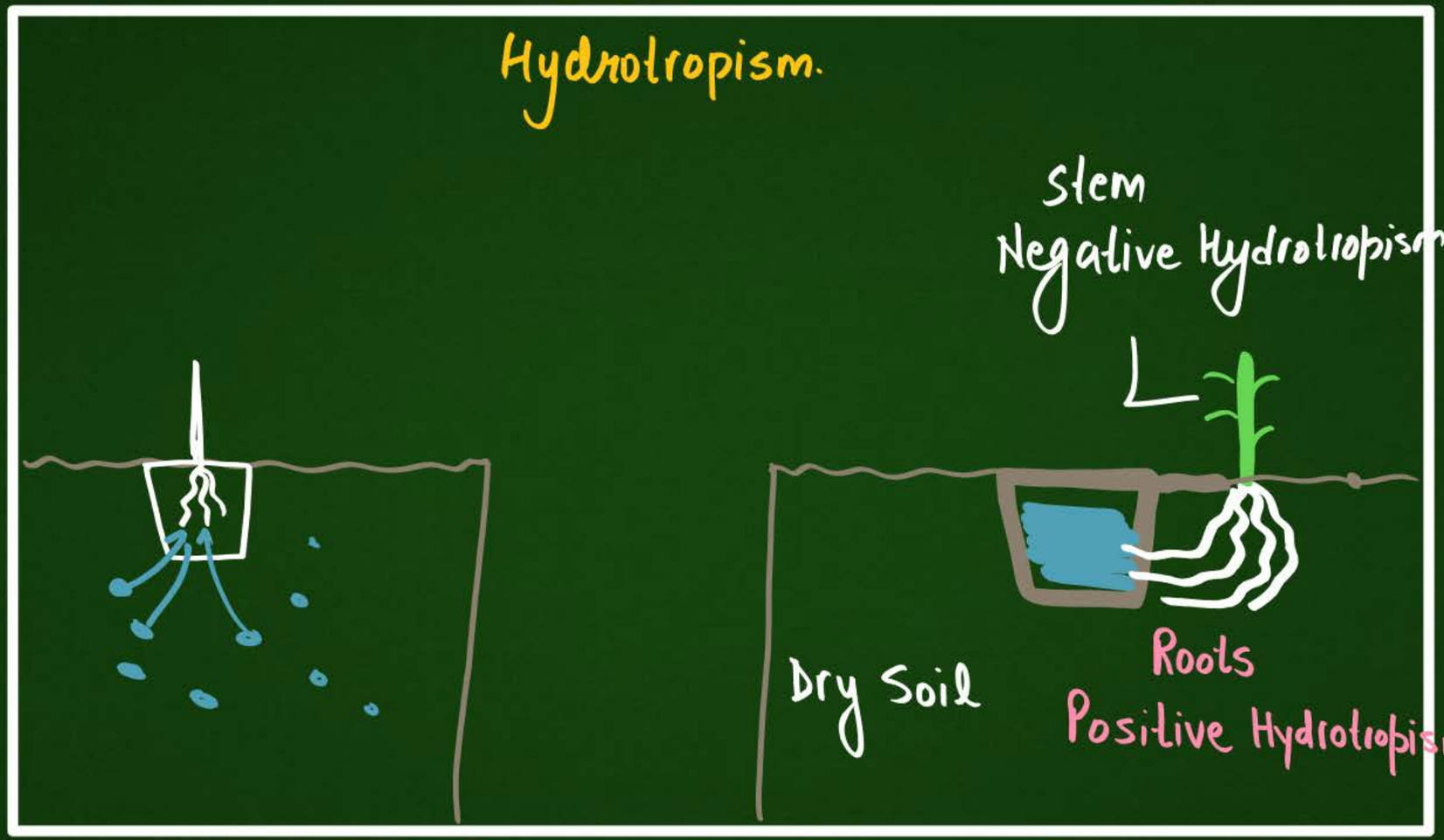


III. Hydrotropism

- Movement of plant parts in response to water or moisture
- When plant parts grow towards source of moisture called positively hydrotropic.
- Roots are positively hydrotropic as water available in soil.
- When plant part grow away from source of moisture is called negatively hydrotropic (shoot)

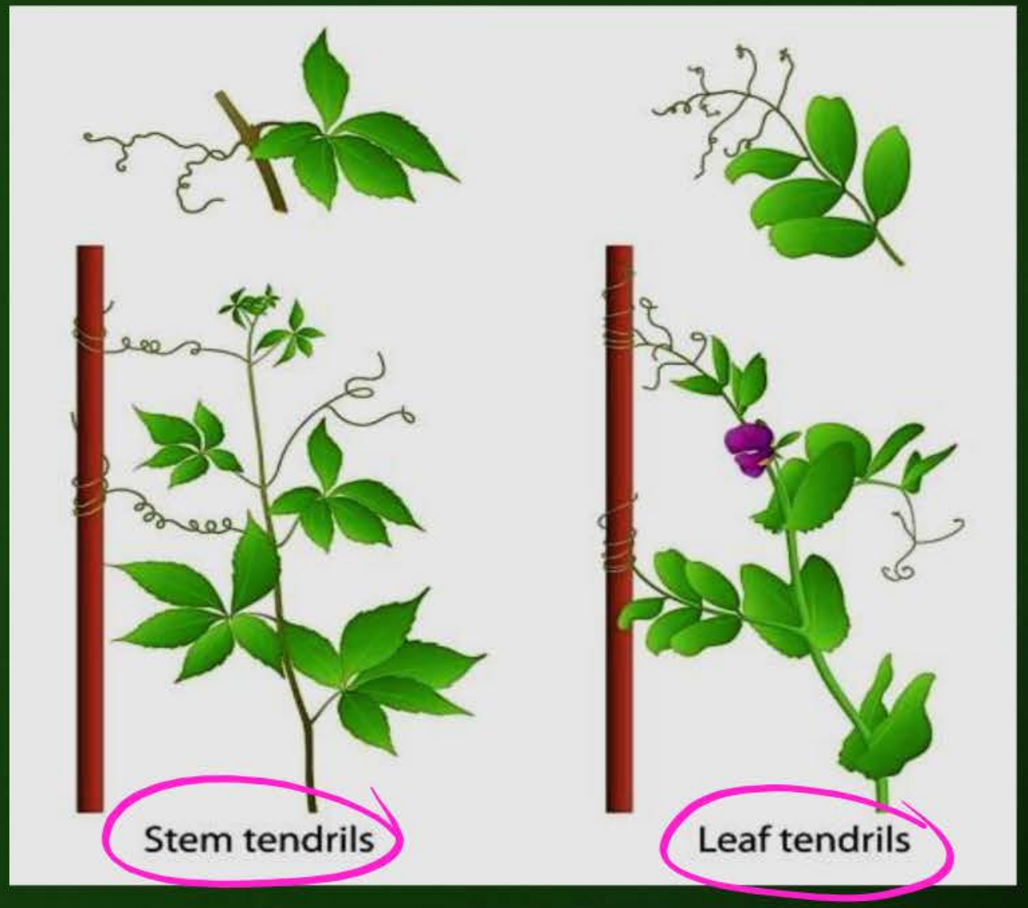


Hydrotropism.



IV. Thigmotropism

- Growth movement of plant parts in response to touch stimulus
- Plants such as sweet peas, cuscuta and vines have tendrils which coil around other plants in response to one sided touch.
- Entire tendril becomes sensitive to stimulus.
- Example of Thigmotropism is tendrils.



Thigmo → touch

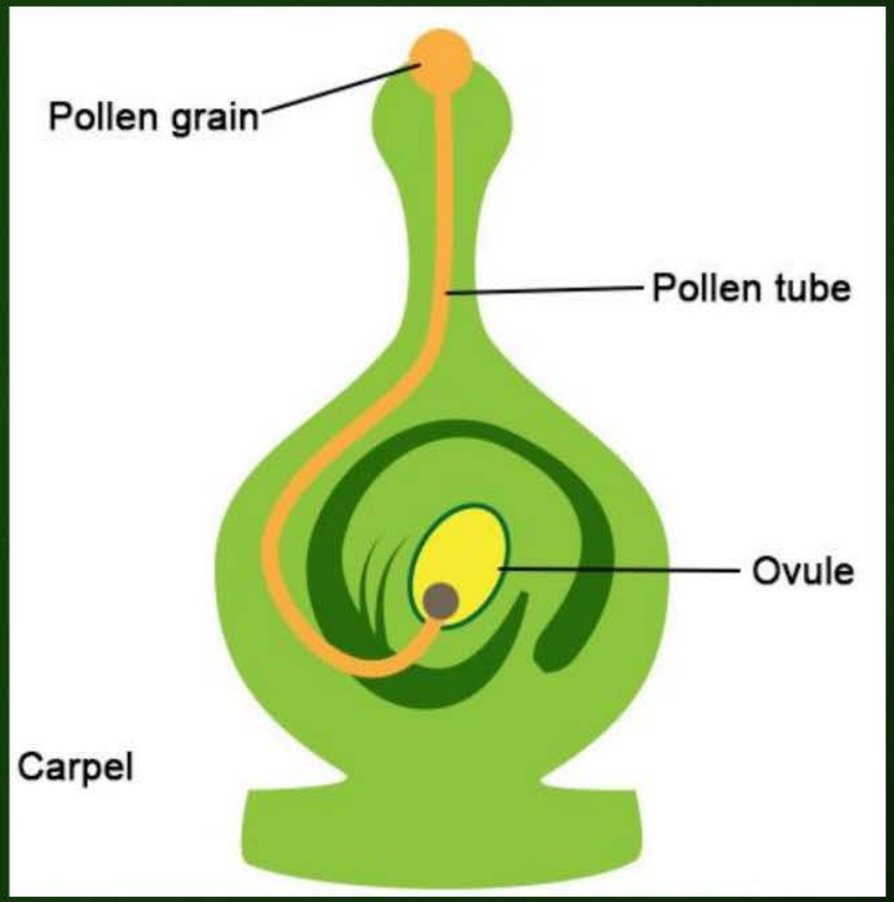
Auxin moved away from touch.

'Cell Elongation.'

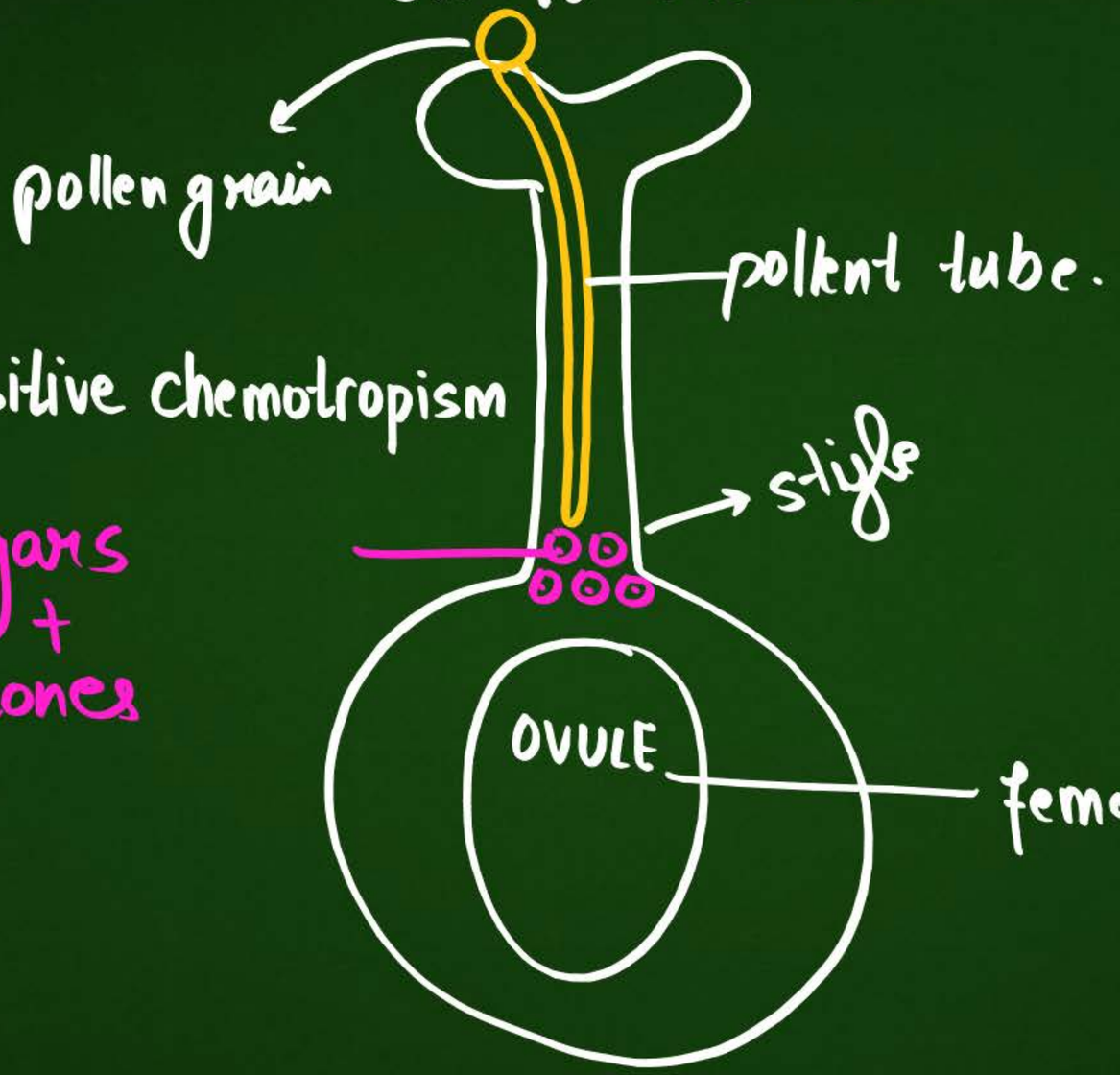


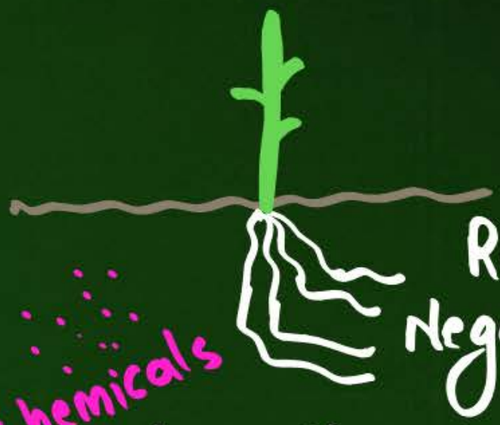
V. Chemotropism

- Growth of plant organs in response to chemicals.
- **Example:** movement of pollen-tube of angiosperm and gymnosperm towards sugars and peptones secreted by neck canal cells of female gametophyte.



Chemo- Chemical





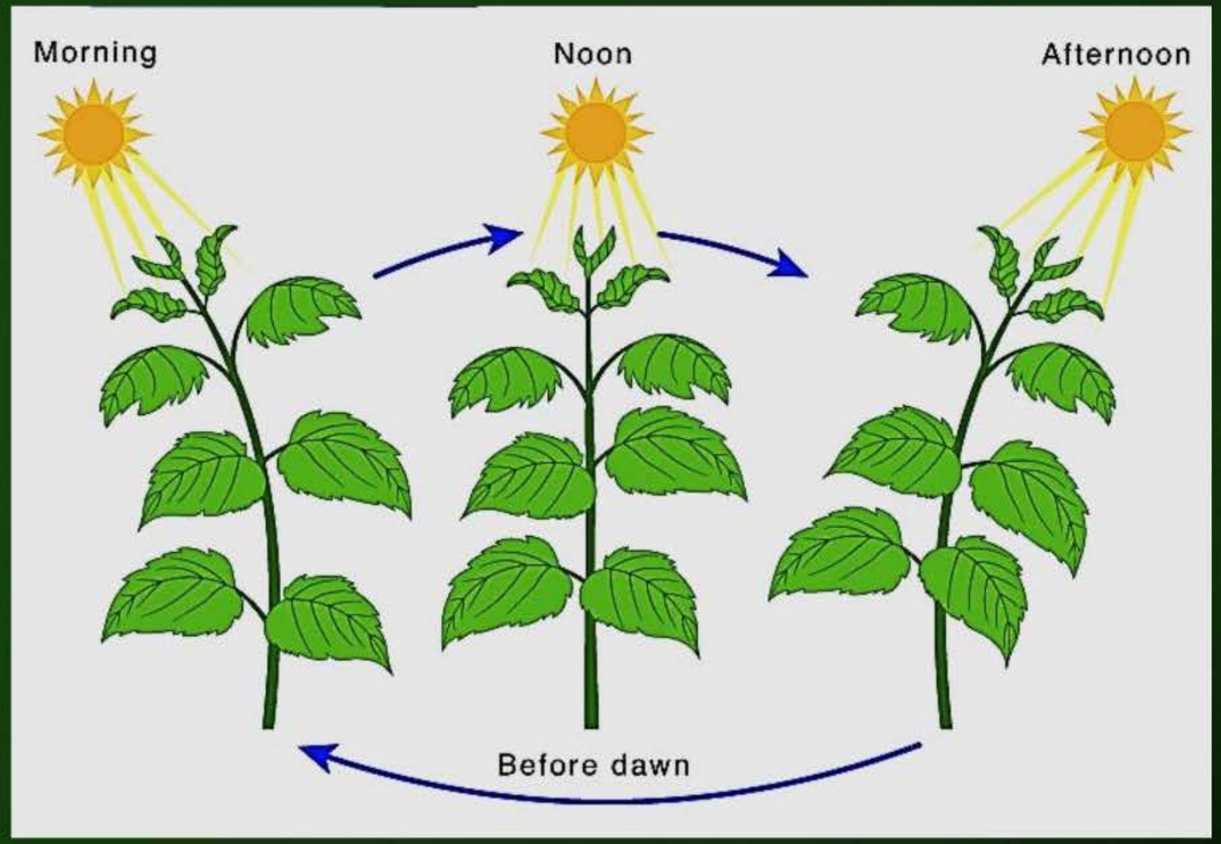
Parthenocarpy

- Natural or artificially induced production of fruit without fertilisation of ovules, which make fruits seedless. Example – banana.

V.imP.

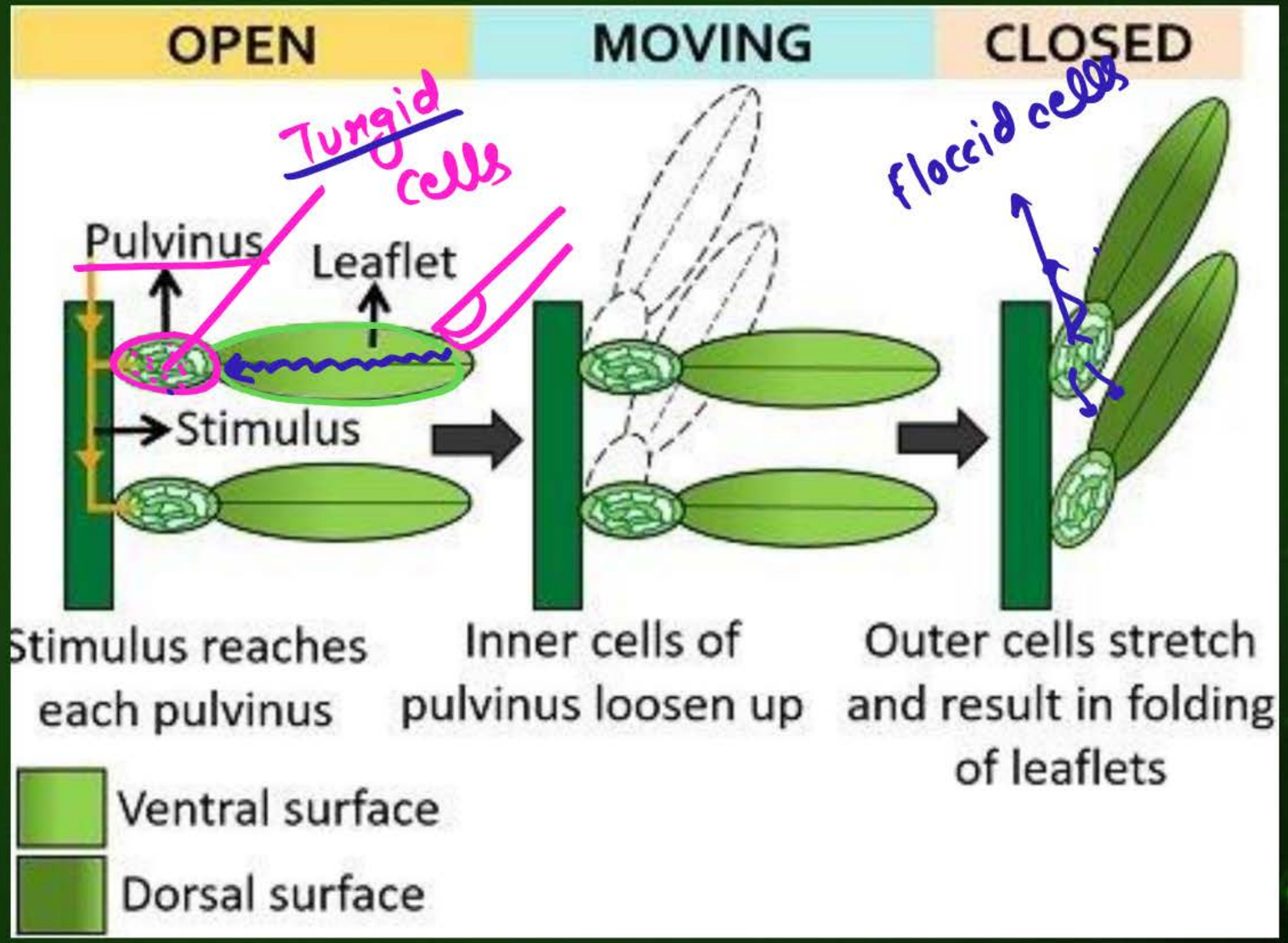
Heliotropism

- Sunflower exhibit heliotropism in which sunflower head moves according to the sun (east to west direction). This is because sunflower contains auxin which is sensitive to sunlight.
- Hence, they migrate from the part of the plant bathed in sunlight to the shaded region of the stem.
- Auxins stimulate the growth of shaded region of stem so that the flower end up bending towards the sun.



Nastic Movement

The movement of plant parts in response to stimulus, but it not directional neither growth related.



QUESTION

Nastic movements differ from tropic movements in being

A Nondirectional

B Movements of variation

C Stimulated by chemicals

D Directional

QUESTION

Pneumatophores show

Mangroove → Normal Roots
↳ Pneumatophores



A thigmotropism

B negative phototropism

C negative geotropism (ageotropism)

D positive geotropism

QUESTION

Fall of mature leaves and fruits from plants is triggered by which of the following substance?

A Auxin

B Cytokinin

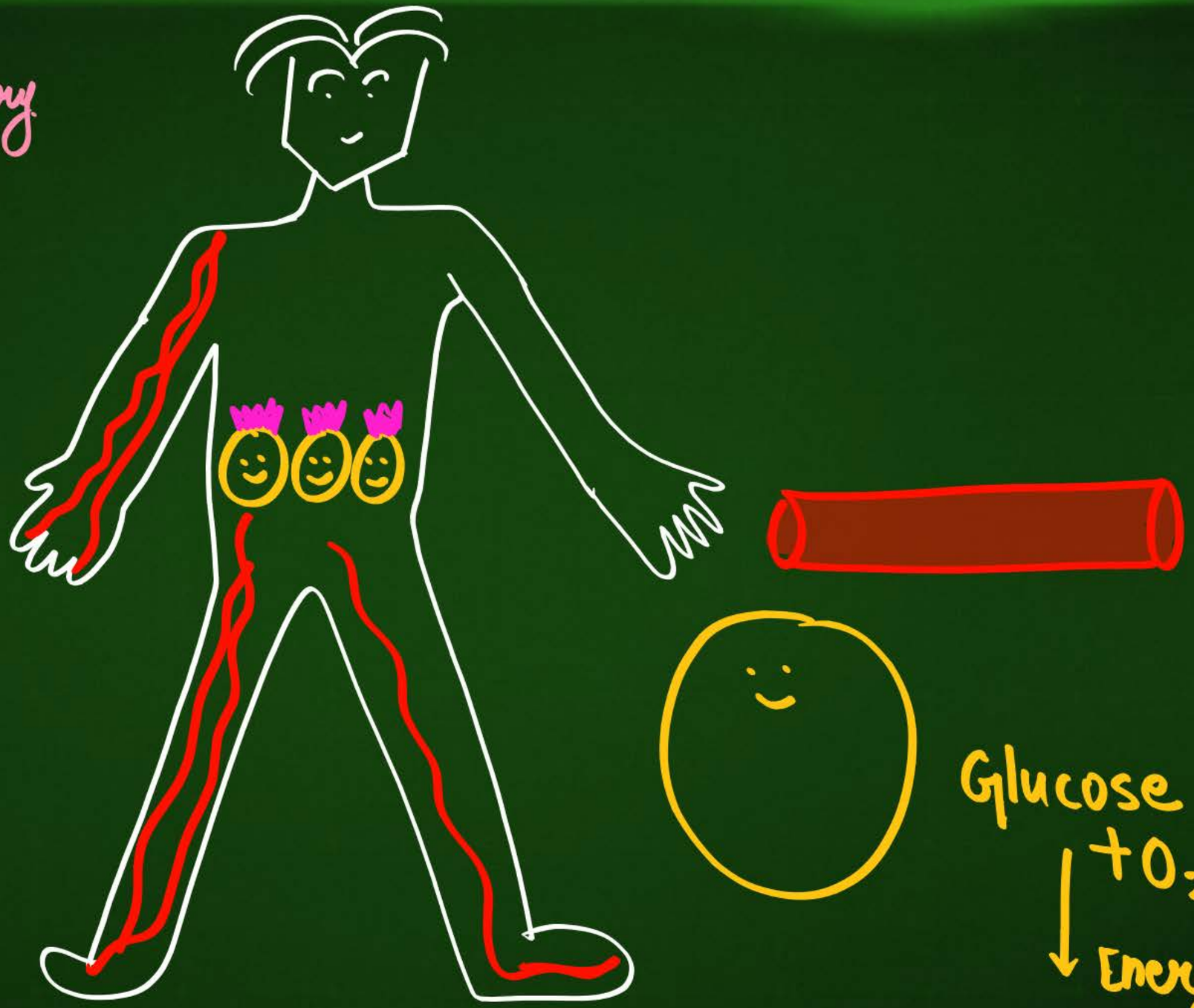
C Gibberellin

D Abscisic acid

The Circulatory System

Components of Circulatory System

- BLOOD
- BLOOD VESSELS
  - Arteries
  - Veins
  - Capillaries
- HEART



Glucose $+O_2$ → Energy $+CO_2$

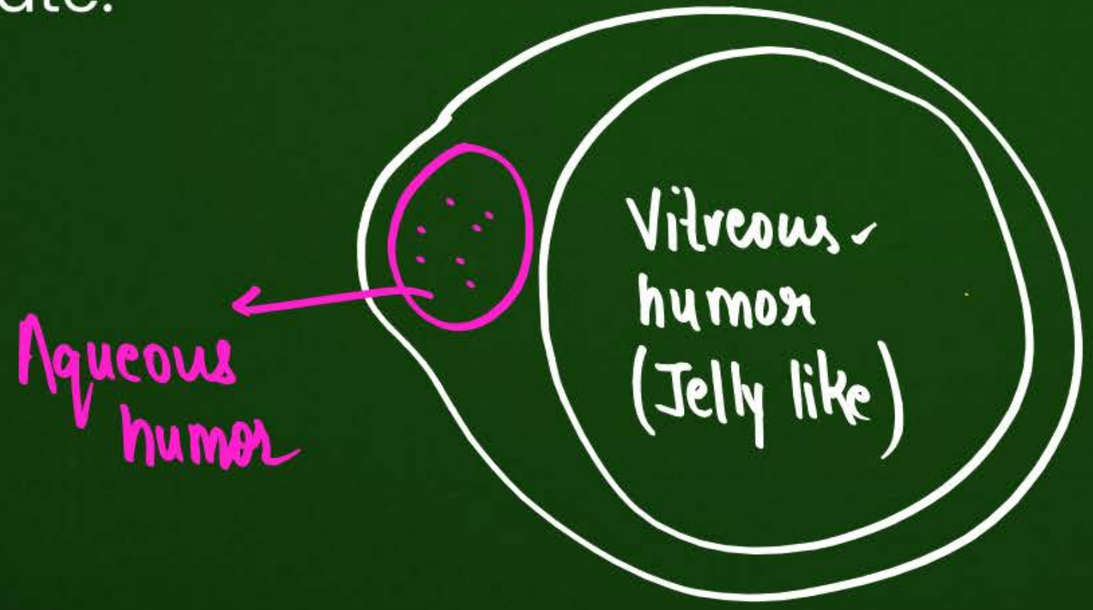
Fluids in Our Body

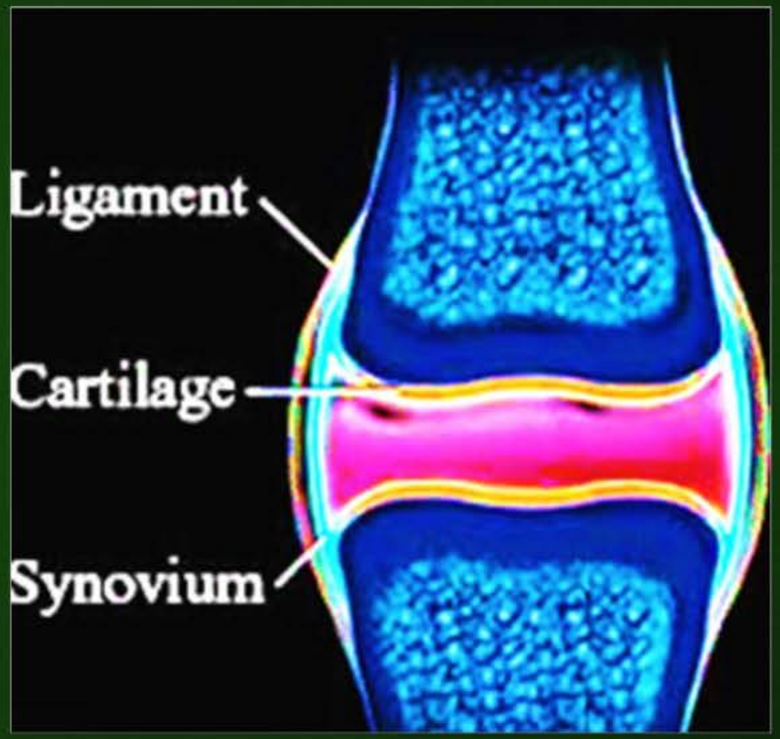
- **Blood**, contained in the heart and in the blood vessels (arteries, veins and capillaries) of the circulatory system.
- **Tissue fluid**, occupying spaces between cells in the organs.
- **Lymph**, which is contained within lymph vessels and lymphatic organs such as the spleen and the tonsils.



Non Circulating Fluids in Our Body

- Some fluids located in particular organs such as synovial fluid filled in the cavities of skeletal joints, vitreous humour in the eye, etc. They do not circulate.





Properties of Blood : The Blood

- **Never Stationary –** Blood is always in motion from the heart to the arteries and back through the veins.
- **Colour –** The blood is a somewhat thick fluid, bright red when taken from an artery or dark red when taken from a vein.
- **Volume –** An average adult human has 5 to 6 litres of blood by volume in his body.
- **Taste – saltish –** The blood is slightly alkaline with a pH of 7.3 to 7.45 (7 is neutral, neither acidic nor alkaline).

Thank You

